

Robustness-Aware LLM Leaderboards

From Diagnosing Ranking Fragility to Actively Repairing It

How worst-case data-dropping audits expose structural fragility in Arena-style evaluation — and a deployable framework to fix it.

PRIMARY Huang, Burt, Hutter & Broderick (2025) · [arXiv:2508.11847](#)

CONTEXT Singh et al. (2025) · [arXiv:2504.20879](#) · The Leaderboard Illusion

Vikhyat Yashvanth Koppal · Lichen Mao · Rygel Ginete

What we'll cover

01

Motivation

Why fragile leaderboards are a real-world risk

02

Chatbot Arena & BT

How pairwise voting becomes a single rank

03

Two Failure Modes

Statistical fragility + system-level bias

04

AMIP Method

Worst-case data-dropping in plain English

05

Our Proposal: RAL

Three components — diagnose, repair, harden

06

Evaluation Plan

What we test on the 140k Arena dataset

Leaderboards drive real decisions

Why ranking position matters

- 👍 Enterprise procurement and vendor selection
- 👍 Research baselines and paper comparisons
- 👍 Government and regulatory model assessment
- 👍 Developer toolchain decisions (which API to call)
- 👍 Marketing claims and competitive positioning

“

When a measure becomes a target, it ceases to be a good measure.” —

Goodhart's Law

Stats: Chatbot Arena (Chiang et al., 2024) · 0.003% finding from Huang et al. (2025)

1M+

Battles collected
on Chatbot Arena

>240

Models ever
benchmarked

0.003%

Of votes can flip
the top-ranked LLM

Chatbot Arena & the Bradley–Terry model

How Chatbot Arena works (Chiang et al., 2024)

- 1 User submits a prompt to two anonymous models
- 2 User reads both responses side-by-side
- 3 User votes: A wins / B wins / Tie
- 4 Vote becomes a pairwise preference record
- 5 Bradley–Terry MLE fit over all collected votes
- 6 Models ranked by estimated BT score β

Bradley–Terry probability (Bradley & Terry, 1952)

$$P(i \text{ beats } j) = \frac{e^{\beta_i}}{e^{\beta_i} + e^{\beta_j}}$$

β estimated by maximum likelihood (MLE) over all votes

Key property — ranking stability depends on β -gaps

- Wide β gap $\rightarrow$ decisive, robust ranking
- Narrow β gap $\rightarrow$ ranking is fragile
- MT-Bench: large gaps (expert prompts)
- Chatbot Arena: tiny gaps at the top

Removing 0.003% of votes flips the #1 LLM

2

preferences dropped

out of 57,477 total — that's 0.003% of the data

Result: the top-ranked model on Chatbot Arena loses the #1 spot

#1 → #2

Top model loses
first place under AMIP

0.003%

Fraction of votes that
determines the top spot

18.1%

Drop fraction needed
on MT-Bench (~6,000× safer)

The Leaderboard Illusion

Five systemic issues from the Singh et al. (2025) audit of ~2M battles, 243 models, 42 providers

1

Private Testing & Selective Disclosure

Meta tested 27 private LLaMA-4 variants pre-release; only the best was kept.

2

Severe Data Asymmetry

Google $\approx$ 19.2% + OpenAI $\approx$ 20.4% of all votes. 83 open-weight models share $\approx$ 29.7%.

3

Arena Overfitting

Up to 112% relative gain on ArenaHard from extra Arena data — Goodhart in action.

4

Silent Model Deprecation

205 of 243 models silently removed — breaks the comparison set's stationarity.

5

Strategic Multi-Submission

Submitting many variants raises the chance one ranks high purely by luck.

How AMIP finds the worst-case drop

AMIP — five steps (Broderick et al., 2020; Huang et al., 2025)

Approximate Maximum Influence Perturbation, extended to BT-based rankings

- 1 Fit BT model on the full vote dataset
- 2 For each vote, compute an influence score (how much the fit moves if removed)
- 3 Sort votes by influence on a target ranking pair
- 4 Greedily remove the top- α highest-influence votes
- 5 Refit BT — confirm whether the predicted rank flip happened

The crucial distinction

Bootstrap CI

“If we redrew votes randomly, how much would the score wobble?”

Assumes IID data · measures sampling variability

AMIP

“Is there any small subset I could delete that reverses the conclusion?”

No IID assumption · measures worst-case fragility

Today's leaderboards: what they miss



What today's leaderboards report

- Bradley–Terry point scores
- Bootstrap confidence intervals (sampling variability)
- Active sampling for information gain
- Battle counts per model



What no platform reports today

- Drop-robustness intervals (worst-case fragility)
- Sampling rules that target fragile pairs
- Aggregators that resist high-leverage votes
- Judge-aware uncertainty estimates

RESEARCH QUESTION

How can preference-based LLM leaderboards be augmented so that ranking fragility is made visible, repairable, and resistant to worst-case data perturbations — without sacrificing scale and openness?

Robustness-Aware Leaderboard (RAL)

A leaderboard column you can read at a glance — fragile vs. robust ranks made visible.

MODEL	BT SCORE	95 % CI	A_FLIP	STATUS
GPT-5	1412	[1405, 1419]	0.0031%	FRAGILE
Claude-4.6	1407	[1399, 1414]	0.41%	MODERATE
Gemini-3	1398	[1389, 1407]	2.8%	ROBUST

← new column: minimum % of votes whose removal would flip this rank



Diagnostic

Report α_{flip} alongside every BT score



Prescriptive

Sample new battles where rankings are fragile



Robust estimation

Cap individual-vote leverage in BT fit

Three components, one closed loop

01

DIAGNOSTIC

Robustness Intervals



- Apply AMIP to each pairwise rank
- Report α_{flip} per adjacent pair
- $O(N)$ cost on top of one BT fit
- Drop-in column on the leaderboard

Extends Broderick et al. (2020); Huang et al. (2025)

02

PRESCRIPTIVE

Influence-Gain Sampling



- Steer next matchup toward fragile pairs
- Generalizes Fisher-information sampling
- Closes the loop: audit $\rightarrow$ action
- λ tunes information $\leftrightarrow$ influence balance

Extends Chiang et al. (2024); Frick et al. (2025)

03

ROBUST

Influence-Capped BT



- Cap top-0.1% high-leverage votes
- Or use Huberized BT loss as alternative
- M-estimator — simple optimizer change
- Reduces fragility before it's reported

Builds on Hunter (2004); Huber-style M-estimators

Evaluation plan on the 140k Arena dataset



Dataset

lmarena-ai/arena-human-preference-140k — ~140,000 pairwise human preferences (Chiang et al., 2024)

C1

Calibration

α_{flip} predicts real flips. Pairs marked robust on training data should rarely flip on held-out votes.

HOW WE TEST

Train/held-out split · predict α_{flip} on train · verify on held-out

C2

Sampling efficiency

Influence-gain sampling reaches a target α_{flip} with fewer new votes than information-gain sampling.

HOW WE TEST

Simulate samplers on 140k · measure votes-to-target curves

C3

Robust aggregation

Influence-capped BT yields rankings with higher α_{flip} — at no cost in held-out predictive log-likelihood.

HOW WE TEST

Refit both estimators · compare log-lik and α_{flip} distributions

All three experiments are runnable on a laptop in under a day · pandas + scipy + statsmodels

CONCLUSION

Two failure modes, one closed-loop fix

FRAGILITY

Statistical fragility is real

On Chatbot Arena, removing 0.003% of votes flips the top model. AMIP catches what bootstrap CIs miss.

Huang et al. (2025)

GOVERNANCE

System bias compounds it

Private testing, data-access asymmetry, and silent deprecation give a few providers structural advantages.

Singh et al. (2025)

OUR DIRECTION

Robustness-Aware Leaderboards

Diagnose with α _flip, sample with influence gain, and fit with influence-capped BT — all on existing infrastructure.

This work

FUTURE WORK

Prompt-level RAL (combine with P2L) • Adversarial robustness vs. vote rigging • Judge heterogeneity & low-rank tensor extensions